Analysis on Manifolds

May 16, 2026

Contents

1 Introduction

1.1 Conservative Fields and Locally Conservative Fields

Definition 1.1 **Locally Conservative**

Let $F : \Omega \to \mathbb{R}^d$ be a vector field. We say that F is **locally conservative** (meshamer mekomit) in Ω if for every $k, \ell = 1, ..., d$,

$$\partial_k F_\ell = \partial_\ell F_k \Longleftrightarrow \frac{\partial F_\ell}{\partial k} = \frac{\partial F_k}{\partial \ell}$$

Definition 1.2 **C^k-smooth**

Domain $\Omega \subset \mathbb{R}^d$ is **C^k-smooth** if for all $x \in \partial\Omega$ there exists a neighborhood $x \in V$ and a C^k-diffeomorphism $\varphi : B_1(0) \to V$ such that $\varphi(B_1(0) \cap \{x_d > 0\}) = V \cap \Omega$ and we call φ a rectifying map.
Informal: Ω is smooth if for all $p \in \partial\Omega$ there exists a neighborhood $p \in V$ such that $V \cap \Omega$ looks like a half circle.

Definition 1.3 **Simply Connected**

X a topological space is **simply connected** if it is a path connected and any loop ($\gamma : [0, 1] \to X$ with $\gamma(0) = \gamma(1)$) in X can be continuously contracted a point (i.e., is homotopic to a constant loop, with no “holes”).

Theorem 1.4

If Ω is simply connected and F is locally conservative in Ω then F is a conservative in Ω.

Proof: In exercise 3. □

Notation

$$\int_\gamma \vec{F}\, \mathrm{d}\vec{\ell} = \int_\gamma F_1\, \mathrm{d}x_1 + F_2\, \mathrm{d}x_2 + ... + F_\ell\, \mathrm{d}x_\ell$$

Exercise 1.5

Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \to V$ a rectifying map with $\mathrm{supp}(F) \subseteq V$ ($\varphi(t, s) = (x(t, s), y(t, s))$).
If $J_p > 0$ then $\varphi(t, 0)$ is a parametrization of $\partial\Omega \cap V$ with positive orientation (since we never use reflection).

Definition 1.6 **Partition of Unity**

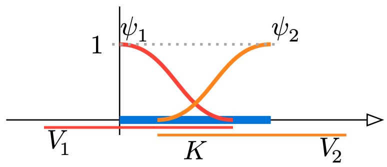
Let $K \subseteq \mathbb{R}^d$ be a compact set and $K \subset V_1 \cup V_2 \cup .... \cup V_m$ a finite open open cover.
Then there exists C^k functions $\psi_1, ..., \psi_m$ defined in a neighborhood of K such that

1. $\forall j, \psi_j \geq 0$
2. $\mathrm{supp}(\psi_j) \subseteq V_k$
3. $\sum_{j=1}^m \psi_j = 1$ on K (meaning for all $x \in K$ we have $\sum_{j=1}^m \psi_j(x) = 1$)

Such collection $\{\psi_j\}_{j=1}^m$ is called **Partition of Unity** on K subordinatory to the cover $\{V_j\}_{j=1}^m$.

Example

Let $K = [0, 2]$, $V_1 = (-1, 1)$, $V_2 = \left(\frac{1}{2}, 3\right)$ so $K \subset U_1 \cup V_2$ and we define ψ_1, ψ_2



Notice that the above, Theorem 1.7 and it’s proof were spread over lectures 6,7,8 and each definition / exercise was in the matching lecture for the part of the proof we used it.

Theorem 1.7 **Green's Theorem**

Let $\Omega \subset \mathbb{R}^2$ be a bounded smooth domain and Let $F = (P, Q)$ be a smooth vector field in a neighborhood of $\partial\Omega$ and $\partial\Omega$ be positively orienated (left leg rule) then

$$\oint_{\partial\Omega} P \,\mathrm{d}x + Q \,\mathrm{d}y = \oint_{\partial\Omega} \vec{F} \,\mathrm{d}\vec{\ell} = \int_{\Omega} \left(\partial_x Q - \partial_y P\right) \mathrm{d}x \,\mathrm{d}y$$

Since $\gamma(t) = (x(t), y(t)), \mathrm{d}x = x'(t) \,\mathrm{d}t, \mathrm{d}y = y'(t) \,\mathrm{d}t$ we get

$$\int_{\gamma} P \,\mathrm{d}x + Q \,\mathrm{d}y = \int (P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)) \,\mathrm{d}t = \int \langle \begin{pmatrix} P \\ Q \end{pmatrix}, \begin{pmatrix} x' \\ y' \end{pmatrix} \rangle \,\mathrm{d}t$$

Remark

Green Theorem works for other domains and for example in the exercise we will prove for rectangles.

Example

Suppose that F is locally conservative in $\Omega \subseteq \mathbb{R}^2$ so $F = (P, Q)$ and $\partial_y P = \partial_x Q$ from Green's theorem

$$\int_{\partial\Omega} \vec{F} \,\mathrm{d}\vec{\ell} = 0$$

Proof of Theorem 1.7:
1st case: Suppose $\mathrm{supp}(F) = \overline{\{x \in \Omega \mid F(x) \neq 0\}} \subset \Omega$.
Since $F|_{\partial\Omega} = 0$ we attain that $\int_{\partial\Omega} \vec{F} \,\mathrm{d}\vec{\ell} = 0$ so we need to show that the integral on the right is also zero.
Since Ω is a bounded smooth domain there exists $M \in \mathbb{R}_+$ so $\Omega \subset [-M, M] \times [-M, M]$ hence

$$\begin{aligned}
\int_{\Omega} \left(\partial_x Q - \partial_y P\right) \mathrm{d}x \,\mathrm{d}y &= \int_{[-M,M]\times[-M,M]} \left(\partial_x Q - \partial_y P\right) \mathrm{d}x \,\mathrm{d}y \\
&= \int_{-M}^{M} \int_{-M}^{M} \partial_x Q \,\mathrm{d}x \,\mathrm{d}y - \int_{-M}^{M} \int_{-M}^{M} \partial_y P \,\mathrm{d}y \,\mathrm{d}x \\
&= \int_{-M}^{M} Q(M, y) - Q(-M, y) \,\mathrm{d}y + \int_{-M}^{M} P(x, M) - P(x, -M) \,\mathrm{d}x = 0
\end{aligned}$$

Where the second equal is Fubini's Theorem and the third one is Fundamental Theorem of Calculus.
2nd case: Assume $\Omega = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1,\ y > 0\}$ and $\mathrm{supp}(F) \subset B_1(0)$.
First, we need to find a parametrization for our boundary: With $-1 \leq t \leq 1$ we have $\gamma(t) = (t, 0)$ with $\gamma'(t) = (1, 0)$ hence

$$\int_{\partial\Omega} P \,\mathrm{d}x + Q \,\mathrm{d}y = \int_{-1}^{1} P(t, 0) \cdot 1 + Q(t, 0) \cdot 0 \,\mathrm{d}t = \int_{-1}^{1} P(t, 0) \,\mathrm{d}t$$

We need to compute $\int \left(\partial_x Q - \partial_y P\right) \mathrm{d}x \,\mathrm{d}y$ over the upper half circle of and show that they are equal.
Over the upper half circle we can replace with rectangle (small enough) and use Fubini's Theorem and Fundamental Theorem of Calculus

$$\int \partial_x Q \,\mathrm{d}x \,\mathrm{d}y = \int_{(-1,1)\times(0,1)} \partial_x Q \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_{-1}^{1} \partial_x Q \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \underbrace{Q(1, y)}_{=0} - \underbrace{Q(-1, y)}_{=0} \,\mathrm{d}y = 0$$

Since we are out of our support. In the same way

$$-\int \partial_y P \,\mathrm{d}x \,\mathrm{d}y = -\int_{(-1,1)\times(0,1)} \partial_y P \,\mathrm{d}x \,\mathrm{d}y = -\int_{-1}^{1} \int_0^1 \partial_y P \,\mathrm{d}y \,\mathrm{d}x = -\int_{-1}^{1} \underbrace{P(x, 1)}_{=0} - P(x, 0)) \,\mathrm{d}x = \int_{-1}^{1} P(x, 0) \,\mathrm{d}x$$

Using a change of variable we finish this case.

End of Lecture 6 – 28/04

3rd case: Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \to V$ a rectifying map with $\operatorname{supp}(F) \subseteq V$ ($\varphi(t,s) = (x(t,s), y(t,s))$).
We can assume that $J_p > 0$ (if not, we can replace φ by reflection – simply flipping one coordinate).
Using Exercise 1.5 since $\operatorname{supp}(F) \subseteq V$ we can write with the definition of line integral

$$\int_{\partial\Omega} P\,\mathrm{d}x + Q\,\mathrm{d}y = \int_{\partial\Omega\cap V} P\,\mathrm{d}x + Q\,\mathrm{d}y = \int_{-1}^{1} P(\varphi(t,0))\frac{\partial x}{\partial t}(t,0) + Q(\varphi(t,0))\frac{\partial y}{\partial t}(t,0)\,\mathrm{d}t$$

For the other side, we start by computing

$$J_\varphi = \det\begin{pmatrix} \frac{\partial x}{\partial t} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial s}\end{pmatrix} = \frac{\partial x}{\partial t}\frac{\partial y}{\partial s} - \frac{\partial x}{\partial s}\frac{\partial y}{\partial t}$$

Since F is smooth we use change of variable to get

$$\begin{aligned}
\int_\Omega \partial_x Q - \partial_y P\,\mathrm{d}x\,\mathrm{d}y &= \int_{\Omega\cap V} \partial_x Q - \partial_y P\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_{\frown} (\partial_x Q\circ\varphi - \partial_y P\circ\varphi) J_\varphi\,\mathrm{d}t\,\mathrm{d}s \\
&= \int_{\frown} (\partial_x Q\circ\varphi - \partial_y P\circ\varphi)\left(\frac{\partial x}{\partial t}\frac{\partial y}{\partial s} - \frac{\partial x}{\partial s}\frac{\partial y}{\partial t}\right)\mathrm{d}t\,\mathrm{d}s \\
&= \int_{\frown} \left((\partial_x Q\circ\varphi)\frac{\partial x}{\partial t} + \cancel{(\partial_y Q\circ\varphi)\frac{\partial y}{\partial t}}\right)\frac{\partial y}{\partial s} - \left((\partial_x Q\circ\varphi)\frac{\partial x}{\partial s} + \cancel{(\partial_y Q\circ\varphi)\frac{\partial y}{\partial s}}\right)\frac{\partial y}{\partial t} \\
&+\left((\partial_x P\circ\varphi)\frac{\partial x}{\partial t} + (\partial_y P\circ\varphi)\frac{\partial y}{\partial t}\right)\frac{\partial x}{\partial s} - \left((\partial_x P\circ\varphi)\frac{\partial x}{\partial s} + (\partial_y P\circ\varphi)\frac{\partial y}{\partial s}\right)\frac{\partial x}{\partial t} \\
&\underset{(1)}{=} \int_{\frown}\left[\frac{\partial(Q\circ\varphi)}{\partial t}\frac{\partial y}{\partial s} - \frac{\partial(Q\circ\varphi)}{\partial s}\frac{\partial y}{\partial t} + \frac{\partial(P\circ\varphi)}{\partial t}\frac{\partial x}{\partial s} - \frac{\partial(P\circ\varphi)}{\partial s}\frac{\partial x}{\partial t}\right]\mathrm{d}t\,\mathrm{d}s \\
&\underset{(2)}{=} \int_{\frown}\left[\frac{\partial}{\partial t}\left((Q\circ\varphi)\frac{\partial y}{\partial s}\right) - \frac{\partial}{\partial s}\left((Q\circ\varphi)\frac{\partial y}{\partial t}\right) + \frac{\partial}{\partial t}\left((P\circ\varphi)\frac{\partial x}{\partial s}\right) - \frac{\partial}{\partial s}\left((P\circ\varphi)\frac{\partial x}{\partial t}\right)\right]\mathrm{d}t\,\mathrm{d}s \\
&= \int_{\frown}\left[\frac{\partial}{\partial t}\left(\underbrace{(P\circ\varphi)\frac{\partial x}{\partial s} + (Q\circ\varphi)\frac{\partial y}{\partial s}}_{\tilde{Q}}\right) - \frac{\partial}{\partial s}\left(\underbrace{(P\circ\varphi)\frac{\partial x}{\partial t} + (Q\circ\varphi)\frac{\partial y}{\partial t}}_{\tilde{P}}\right)\right]\mathrm{d}t\,\mathrm{d}s
\end{aligned}$$

Where (1) is Chain Rule and (2) Leibniz Integral Rule (Notice the line break with the $+$).
If we take $\vec{G}(t,s) = (\tilde{P}, \tilde{Q})$ and since $\gamma(t) = (t,0)$, $\dot{\gamma}(t) = (1,0)$, the 2nd case gives us

$$\int_{\partial\Omega} \tilde{P}\,\mathrm{d}t + \tilde{Q}\,\mathrm{d} = \int_{\frown} \frac{\partial\tilde{Q}}{\partial t} - \frac{\partial\tilde{P}}{\partial s}\,\mathrm{d}t\,\mathrm{d} \Longrightarrow \int \tilde{P}\,\mathrm{d}t + \tilde{Q}\,\mathrm{d}s = \int_{-1}^{1}\tilde{P}\,\mathrm{d}t = \int_{-1}^{1}\left[(P\circ\varphi)\frac{\partial y}{\partial t}(t,0) + (Q\circ\varphi)\frac{\partial y}{\partial t}(t,0)\right]\mathrm{d}ts$$

Which is indeed the left side.
In other words for the 3rd case: we pull our vector field back to the upper half circle and go back to the 2nd case.

End of Lecture 7 – 30/04

4rd case: For all $p \in \partial\Omega$ there exists a neighborhood V_p and a rectifying map $\varphi_p : B_1(0) \to V_p$.
Since $\partial\Omega \subset \bigcup_{p\in\partial\Omega} V_p$ and since $\partial\Omega$ is compact then there exists a finite open subcover $\partial\Omega \subset V_{P_1} \cup \dots \cup V_{p_m}$. Let $V_0 = \Omega$ and $\overline{\Omega} \subset V_0 \cup V_{p_1} \cup \ldots \cup V_{p_m}$ and for simplicity of notations let us write $v_j = v_{\gamma_j}$.
There exists a partition of unity on $\overline{\Omega}$ subordinate to $\{V_j\}_{j=0}^{m}$.
Define $\vec{F}_j := \psi_j \cdot \vec{F}$ and $\operatorname{supp}(\vec{F}_j) \subset V_j$ and we have

$$\sum_{j=1}^{m} \vec{F}_j = \sum_{j=1}^{m}\left(\psi_j \cdot \vec{F}\right) = \left(\sum_{j=1}^{m}\psi_j\right)\cdot\vec{F} \underset{\text{on } \overline{\Omega}}{=} 1\cdot\vec{F} = \vec{F}$$

From the 1st case if we set $\vec{F}_j = (P_j, Q_j)$ since $\operatorname{supp}\left(\vec{F}_0\right) \subseteq \Omega)$ we have that

$$\int_{\partial\Omega} \vec{F}_0 \, \mathrm{d}\vec{\ell} = \int_\gamma \partial_X Q_0 - \partial_y P_0 \, \mathrm{d}x \, \mathrm{d}y$$

By the 3rd case we know that for each $1 \leq j \leq m$ we have

$$\int_{\partial\Omega} \vec{F}_j \, \mathrm{d}\vec{\ell} = \int_\gamma \partial_x Q_j - \partial_y P_j \, \mathrm{d}x \, \mathrm{d}y$$

Summention gives us

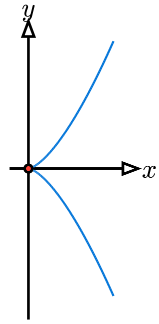
$$\sum_{j=0}^{m} \int_{\partial\Omega} \vec{F}_j \, \mathrm{d}\vec{\ell} = \sum_{j=0}^{m} \int_\gamma \partial_x Q_j - \partial_y P_j \, \mathrm{d}x \, \mathrm{d}y \Longrightarrow \int_{\partial\Omega} \vec{F} \, \mathrm{d}\vec{\ell} = \int_\gamma \partial_x Q - \partial_y P \, \mathrm{d}x \, \mathrm{d}y$$

□

Proof of the existence of partition of unity: TBD. □

2 Smooth Manifolds in $\mathbb{R}^d$

We start with a warning: do not confuse a smooth curve with a smooth parametrization.
For example, take $\gamma(t) = (t^2, t^3)$ with $-\infty < t < \infty$.
This is a smooth parametrization curve but not a smooth curve since $\dot{\gamma}(0) = (0, 0)$.



Definition 2.1 — **Smooth manifolds in $\mathbb{R}^d$**

Let $k \leq d$, a set $M \subseteq \mathbb{R}^d$ is called a **manifold** of dimension k if for all $p \in M$ there exists $W \subseteq \mathbb{R}^d$ a neighborhood of p ($p \in W$), an open set $U \subseteq \mathbb{R}^k$ and a smooth map $\varphi : U \to W$ so that

1. φ is a homeomorphism on it's image
2. $\varphi(U) = M \cap W$
3. $\operatorname{rank}(D\varphi_u) = k$ for all $u \in U$

Informly, a manifolds of dimension k is any topological space U so that any point $p \in U$ has a neighborhood W which is a homeomorphism to a ball in $\mathbb{R}^k$ (A k-dimensional manifold is a space that locally looks like $\mathbb{R}^k$).

End of Lecture 8 – 05/05

TBD – Lecture 9.

End of Lecture 9 – 07/05

Definition 2.2 — **Graph of dim k**

A set $\Gamma \subseteq \mathbb{R}^d$ is **a graph of dim k** if there exists a partition $\{1, 2, ..., d\} = I \uplus J$ with $|I| = k$, an open set $U \subseteq \mathbb{R}^k$ and a function $f : U \to \mathbb{R}^{d-k}$ such that

$$\Gamma = \{x \in \mathbb{R}^d \mid x_I \in U,\ x_J = f(x_I)\}$$

Example

Let $U \subseteq \mathbb{R}^d$ be an open set with $k = d$. We show that U is a graph.
We take $I = \{1, 2, ..., d\}$ and $J = \emptyset$ with $f : U \to \mathbb{R}^0 = \{0\}$ defined by $f(u) = 0$.
Then

$$U = \{x \in \mathbb{R}^d \mid x \in U\} = \{x \in \mathbb{R}^d \mid x_I \in U, x_J = f(x_I)\}$$

Since $J = \emptyset$, the condition $x_J = f(x_I)$ is vacuous.

Definition 2.3 — **Smooth manifolds of dim k by graph**

A set $M \subseteq \mathbb{R}^d$ is **a smooth manifold of dim k** if for every $p \in M$ there exists an open neighborhood $W \subseteq \mathbb{R}^d$ of p such that $W \cap M$ is a graph of dim k of a smooth function. We say that M is locally a smooth graph of dim k for a $p \in M$.

Definition 2.4 — **Smooth manifolds of dim k**

Let $k \leq d$. A set $M \subseteq \mathbb{R}^d$ is called **a smooth manifold of dim k** if for every $p \in M$ there exists an open neighborhood $W \subseteq \mathbb{R}^d$ of p, an open set $U \subseteq \mathbb{R}^k$ and a smooth map $\varphi : U \to W$ such that

1. $\varphi : U \to \varphi(U)$ is a homeomorphism
2. $\varphi(U) = M \cap W$
3. $\operatorname{rank}(D\varphi_u) = k$ for all $u \in U$

Example

We look at the curve

$$\gamma(t) = (\sin(t), \sin(2t))$$

We define $S = \left\{\gamma(t) \mid 0 < t < \frac{5\pi}{4}\right\}$ and ask is S a smooth manifold according to the definition with graphs?
Observe that

$$\gamma(0) = \gamma(\pi) = (0, 0)$$

hence the parametrization is not injective.
On the other hand, define $U = \left(-\varepsilon, \frac{5\pi}{4}\right)$ then

$$(D\varphi)|_{t=0} = (\cos(t), 2\cos(2t))|_{t=0} = (1, 2)$$

has full rank.
However, since $\varphi(0) = \varphi(\pi)$, the map φ is not injective, hence it is not a homeomorphism onto its image.
Therefore this parametrization does not define a smooth manifold chart.
In fact, near $(0, 0)$ the set looks like two curves crossing, so S is not a smooth manifold.

Proposition 2.5

Every smooth manifold by the graph definition is a smooth manifold.

Proof: Let

$$\Gamma = \left\{x \in \mathbb{R}^d \mid x_I \in U,\ x_J = f(x_I)\right\}$$

where $f : U \to \mathbb{R}^{d-k}$, $\{1, 2, ..., d\} = I \cup J$ and $|I| = k$.
Define $\varphi : U \to \Gamma$ by

$$(\pi_I \circ \varphi)(u) = u, \quad (\pi_J \circ \varphi)(u) = f(u)$$

The map φ is smooth since each coordinate is a smooth function.
Moreover, the Jacobian matrix of φ contains the identity matrix I_k as a block, hence $\operatorname{rank}(D\varphi_u) = k$ for all $u \in U$. Hence

$$\varphi(U) = \{\varphi(u) \mid u \in U\} = \left\{x \in \mathbb{R}^d \mid (\pi_I \circ \varphi)(x) = u,\ (\pi_J \circ \varphi)(x) = f(u)\right\}$$

We have $\varphi : U \to \Gamma$ invertible since

$$(\pi_I \circ \varphi)(u) = \pi_I(\varphi(u)) = u$$

Moreover,

$$\pi_I\left(\varphi\left((\pi_I \circ \varphi)^{-1}(\pi_I(x))\right)\right) = \pi_I(x)$$

and

$$\pi_J(\varphi\left((\pi_I \circ \varphi)^{-1}(\pi_I(x))\right) = f(\pi_I(x)) = \pi_J(x)$$

Therefore

$$\varphi\left((\pi_I \circ \varphi)^{-1}(\pi_I(x))\right) = x$$

So $\pi_I : \Gamma \to U$ is continuous and is the inverse of $\varphi : U \to \Gamma$. □

Theorem 2.6

Every smooth manifold is a graph manifold.

Proof: Let $p \in M$, $p \in W \subseteq \mathbb{R}^d$ open, $U \subseteq \mathbb{R}^k$ and $\varphi : U \to W$ with $\varphi(U) = W \cap M$, such that $(D\varphi)|_{\varphi^{-1}(p)}$ is of full rank.
There exists $I \subseteq \{1, 2, ..., d\}$ with $|I| = k$ so that $(D\varphi)|_{I \times \{1,2,...,k\}}$ is invertible.
We look at the map $\pi_I \circ \varphi : U \to \mathbb{R}^k$ which is a smooth map from U to $\mathbb{R}^k$ and

$$D(\pi_I \circ \varphi)|_{\varphi^{-1}(p)} = (D\varphi)|_{I \times \{1,2,\dots,k\}}$$

Is invertible.
By the inverse function theorem, $\pi_I \circ \varphi$ is a local diffeomorphism at $\varphi^{-1}(p)$, so there exists $\varphi^{-1}(p) \in U_0 \subset U$ open and there exists $V_0 \subseteq \mathbb{R}^k$ open so that $(\pi_I \circ \varphi) : U_0 \to V_0$ is a diffeomorphism.
We define $f : V_0 \to \mathbb{R}^{d-k}$ by

$$f(v) = \pi_J\Big(\varphi\Big((\pi_I \circ \varphi)^{-1}(v)\Big)\Big)$$

Where $J = \{1, 2, \dots, d\} \setminus I$. Let W_0 be such that $\varphi(U_0) = W_0 \cap M$.
We need to verify that

$$\underbrace{W_0 \cap M}_{=A} = \underbrace{\{x \in \mathbb{R}^d \mid \pi_I(x) \in V_0,\ \pi_J(x) = f(\pi_I(x))\}}_{=B}$$

Since $W_0 \cap M = \varphi(U_0)$, let $u \in U_0$.
Then there exists $v \in V_0$ such that $(\pi_I \circ \varphi)^{-1}(v) = u$, we aim to show that $\varphi(u) \in B$.
From our definition, $\pi_I(\varphi(u)) = v \in V_0$.
Also,

$$\pi_J(\varphi(u)) = \pi_J\Big(\varphi\Big((\pi_I \circ \varphi)^{-1}(v)\Big)\Big) = f(v)$$

Hence $\varphi(u) \in B$.
Why is $B \subseteq A$? Let $x \in B$ and define

$$u = (\pi_I \circ \varphi)^{-1}(\pi_I(x)) \in U_0$$

Then

$$\pi_I(\varphi(u)) = (\pi_I \circ \varphi)(u) = \pi_I(x)$$

Also,

$$\pi_J(\varphi(u)) = f(\pi_I(\varphi(u))) = f(\pi_I(x)) = \pi_J(x)$$

Because the coordinates agree on both I and J, we conclude that $\varphi(u) = x$, hence $x \in A$. $\square$

2.1 Transition Maps

Let M be a smooth manifold of dim k and suppose we have

$$\varphi_1 : U_1 \to W_1, \quad \varphi_2 : U_2 \to W_2$$

where W_1, W_2 are neighborhoods of p.
Denote

$$\tilde{U}_1 = \varphi_1^{-1}(W_1 \cap W_2 \cap M), \quad \tilde{U}_2 = \varphi_2^{-1}(W_1 \cap W_2 \cap M)$$

We define $\tau_{12} : \tilde{U}_1 \to \tilde{U}_2$ by

$$\tau_{12}(u) = \varphi_2^{-1}(\varphi_1(u))$$

where $\varphi_2 : U_2 \to W_2 \cap M$.
We ask ourselves: is τ_{12} a smooth map?
After possibly shrinking neighborhoods, we may assume both charts are represented as graphs over the same coordinate subset I.
Then

$$\tau_{12} = \varphi_2^{-1} \circ \varphi_1 = (\pi_I \circ \varphi_2)^{-1} \circ (\pi_I \circ \varphi_1)$$

Since compositions and inverses of smooth maps are smooth, the transition map is smooth.

End of Lecture 10 – 12/05

> **Remark**
>
> Copy another definition and equality from restriction 5 + exercise 5.